

COMPLETE PROJECT RECORD

SOC HOME LAB

Complete Technical Documentation

Phases 0–10 | Build, Attack, Detect, Investigate, Harden, Validate

Environment: Isolated VirtualBox SOC Home Lab

Domain: soclab.local

Primary systems: DC01 / Splunk, WIN11-01, KALI-01

Network: 192.168.10.0/24 Host-Only

Documentation date: 12 July 2026

Project status: Core technical objectives completed

OUTCOME

The lab demonstrates an end-to-end defensive workflow: infrastructure deployment, endpoint telemetry, attack simulation, SIEM detection, incident investigation, security hardening, and post-hardening validation.

Document Purpose and Scope

This document consolidates the complete SOC Home Lab project from Phase 0 through Phase 10. It records the implemented architecture, work performed, validation methods, incident findings, hardening controls, troubleshooting decisions, evidence, and remaining improvement opportunities.

The report distinguishes implemented work from the original blueprint. Where the final configuration differed from the initial plan, the working configuration is documented as the authoritative state.

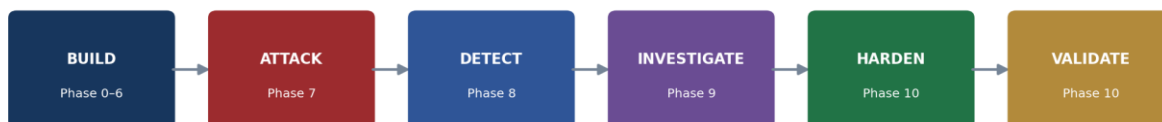
EVIDENCE BOUNDARY

Most Phase 0–8 screenshots are referenced by filename in the exported conversation. Phase 9 and Phase 10 were directly verified from their submitted evidence packages and final reports.

Executive Summary

The project created an isolated Windows-focused SOC environment using VirtualBox. DC01 provided Active Directory, DNS, and Splunk services; WIN11-01 acted as the monitored endpoint; and KALI-01 generated authorized test activity. Windows Security logs and Sysmon telemetry were forwarded into Splunk for search, dashboards, alerts, and investigation.

The principal incident scenario was an SMB credential attack against WIN11-01. Splunk recorded 31 failed logons followed by four successful localadmin network logons from 192.168.10.30. Phase 9 reconstructed and documented the event. Phase 10 then enabled account lockout, verified host and protocol protections, repeated the test, and demonstrated that Windows returned STATUS_ACCOUNT_LOCKED_OUT while monitoring remained operational.



Complete defensive lifecycle demonstrated in the SOC Home Lab

Figure 1. Defensive lifecycle demonstrated across Phases 0–10.

Project Status at a Glance

Phase	Focus	Status	Primary result
0	Preparation and network	Complete	VirtualBox and isolated host-only network prepared
1	Windows Server VM	Complete	DC01 deployed and stabilized
2	AD DS and DNS	Complete	soclab.local operational
3	Windows client	Complete	WIN11-01 joined and used in the domain
4	Splunk platform	Complete	Splunk available on DC01 and receiving data
5	Forwarder and Sysmon	Complete	Endpoint telemetry collected
6	Kali attacker	Complete	KALI-01 configured at 192.168.10.30
7	Safe simulations	Core complete	Nmap and NetExec scenarios executed
8	Detections and dashboard	Complete	Queries, alert, and dashboard operational
9	Investigation	Complete	High-severity credential incident documented
10	Hardening and validation	Complete	Account lockout enforced; validation passed

Implemented Architecture



VirtualBox Host-Only Network • 192.168.10.0/24 • DHCP disabled • Isolated lab traffic

Figure 2. Implemented SOC Home Lab architecture and data flow.

Role	System	Address	Primary functions
Host / isolation	VirtualBox host adapter	192.168.10.1	Host-only network; DHCP disabled
Domain and SIEM	DC01 / Splunk	192.168.10.10	AD DS, DNS, Splunk Enterprise, TCP 9997 receiver
Monitored endpoint	WIN11-01	192.168.10.20	Windows client, Security logs, Sysmon, Universal Forwarder
Authorized attacker	KALI-01	192.168.10.30	Nmap reconnaissance and NetExec SMB testing

Telemetry Flow

1. Windows generates Security, System, Application, and Sysmon Operational events.
2. The Splunk Universal Forwarder on WIN11-01 collects enabled event channels.
3. Telemetry is sent to the Splunk receiver at 192.168.10.10 on TCP port 9997.
4. Splunk indexes, searches, visualizes, and alerts on the collected activity.
5. Analyst searches correlate attacker IP, account, host, event ID, and time.

IMPLEMENTED CONFIGURATION

The original blueprint proposed separate wineventlog and sysmon indexes. The working deployment used index=main. Sysmon arrived as XML with renderXml=true, so required fields were extracted from _raw during investigation.

PHASE 0

Phase 0 — Lab Preparation

Objective

Prepare a safe, organized virtualization environment before deploying any systems.

Work Performed

- Installed Oracle VirtualBox and confirmed that virtualization could support the required virtual machines.
- Created VirtualBox Host-Only Ethernet Adapter #2 with address 192.168.10.1/24.
- Disabled DHCP so each lab system could use a predictable static address.
- Downloaded the Windows Server 2022, Windows 11, Ubuntu Server, and Kali Linux installation or VM media during initial planning.
- Created organized locations for ISOs, VM files, screenshots, reports, configurations, and Splunk searches.
- Adopted a checkpoint approach: verify networking, services, and logs before advancing to the next phase.

Validation and Result

- The isolated 192.168.10.0/24 network supported later communication among DC01, WIN11-01, and KALI-01.
- The attack simulations remained within the Host-Only network rather than using a bridged adapter.
- Project evidence was saved throughout later phases, confirming that the folder and screenshot workflow was used.

Recorded Evidence

- Host-Only adapter configuration screenshots referenced in the conversation.
- Organized screenshot and report folders used in Phases 6–10.

**SAFETY
DECISION**

Host-Only networking was intentionally used to keep authorized attack traffic isolated from external networks.

PHASE 1

Phase 1 — Windows Server 2022 and DC01

Objective

Deploy the Windows Server virtual machine that would become the central domain and monitoring system.

Work Performed

- Created the DC01 VM in VirtualBox using Windows Server 2022 Evaluation media.
- Allocated a dynamically expanding VDI disk and appropriate processor and memory resources for the lab.
- Configured the VM storage, boot order, clipboard integration, and Host-Only network adapter.
- Installed Windows Server, completed administrator setup, and applied the DC01 hostname.
- Troubleshoot Guest Additions, boot order, network-adapter selection, and VM performance while moving VM storage to SSD.

Validation and Result

- DC01 remained operational throughout the project and hosted the services required by later phases.
- WIN11-01 forwarded telemetry to 192.168.10.10:9997, confirming DC01 network reachability.
- Splunk searches were later performed locally on DC01 through port 8000.

Recorded Evidence

- DC01 installation, configuration, network, and VirtualBox screenshots referenced in the conversation.

PHASE 2

Phase 2 — Active Directory Domain Services and DNS

Objective

Create the soclab.local identity environment and provide centralized naming and authentication.

Work Performed

- Configured DC01 with the static address 192.168.10.10.
- Installed and promoted Active Directory Domain Services.
- Created the soclab.local domain and enabled DNS services on the domain controller.
- Created and used domain identity records required for Windows client testing.
- Verified domain operation indirectly through later client authentication, domain policy queries, and Phase 10 lockout-policy changes.

Validation and Result

- Get-ADDefaultDomainPasswordPolicy returned the active domain policy during Phase 10.
- The domain lockout threshold was successfully changed from 0 to five attempts.
- WIN11-01 operated within the domain environment and domain credentials were used during earlier checkpoints.

Recorded Evidence

- Active Directory, DNS, user, and domain-policy screenshots referenced across the conversation and Phase 10 evidence.

**EVIDENCE
IMPROVEMENT**

For a public portfolio, add one clear screenshot of nslookup soclab.local and one view of the final OU/user structure.

PHASE 3

Phase 3 — Windows 11 Client Deployment

Objective

Deploy the monitored Windows endpoint and integrate it with the domain and lab network.

Work Performed

- Created and installed the WIN11-01 virtual machine.
- Assigned the static address 192.168.10.20 and used DC01 as the domain/DNS service.
- Connected the VM to the same Host-Only network used by DC01.
- Joined the Windows client to soclab.local and used a domain user account during validation.
- Moved VM storage to SSD to resolve performance problems while preserving the working VM configuration.

Validation and Result

- WIN11-01 communicated with DC01 and KALI-01 during later telemetry and attack tests.
- Windows Security events and Sysmon events from WIN11-01 appeared in Splunk.
- The client processed network logons and generated Events 4624, 4625, and 4740.

Recorded Evidence

- WIN11-01 network, domain, account, and VirtualBox screenshots referenced in the exported conversation.

PHASE 4

Phase 4 — Splunk Enterprise Deployment

Objective

Provide a central SIEM platform for receiving, searching, visualizing, and alerting on Windows telemetry.

Work Performed

- Deployed Splunk Enterprise on the implemented DC01/Splunk system at 192.168.10.10.
- Used the Splunk web interface on port 8000.
- Enabled the Splunk receiving service on TCP port 9997.
- Used index=main for the final working deployment.
- Created searches for Windows authentication, Sysmon telemetry, Nmap-related activity, and brute-force behavior.

Validation and Result

- WIN11-01 successfully forwarded Security and Sysmon data to the Splunk receiver.
- Splunk supported the Phase 9 incident investigation and the Phase 10 validation searches.
- The SOC dashboard and Brute Force Detection alert remained operational after hardening.

Recorded Evidence

- Splunk search, receiver, index, alert, and dashboard screenshots referenced in the conversation and evidence packages.

**ARCHITECTURE
DECISION**

Although Ubuntu Server appeared in the initial download plan, the verified implemented topology used DC01 / Splunk at 192.168.10.10.

PHASE 5

Phase 5 — Universal Forwarder and Sysmon

Objective

Collect native Windows audit events and enhanced endpoint telemetry from WIN11-01.

Work Performed

- Installed the Splunk Universal Forwarder on the Windows endpoint.
- Configured Windows Security, System, Application, and Sysmon Operational event collection.
- Installed Sysmon and enabled endpoint telemetry including process creation, file creation, and network connection events.
- Configured the Sysmon channel with renderXml=true in the working inputs.conf.
- Validated that the SplunkForwarder and Sysmon64 services were running.

Validation and Result

- Local Get-WinEvent checks confirmed that Sysmon Event ID 3 existed on WIN11-01.
- Restarting SplunkForwarder reloaded the input and restored fresh Sysmon ingestion.
- Splunk received 238 fresh Sysmon events during recovery validation.
- XML extraction produced Image, SourceIp, SourcePort, DestinationIp, and DestinationPort fields.

Recorded Evidence

- Phase 9 Figures 9–11 and Phase 10 service and live-event screenshots.

TROUBLESHOOTING RESULT

Sysmon itself was functioning. The zero-result problem was in collection/reload and XML field extraction, not in the local Sysmon NetworkConnect configuration.

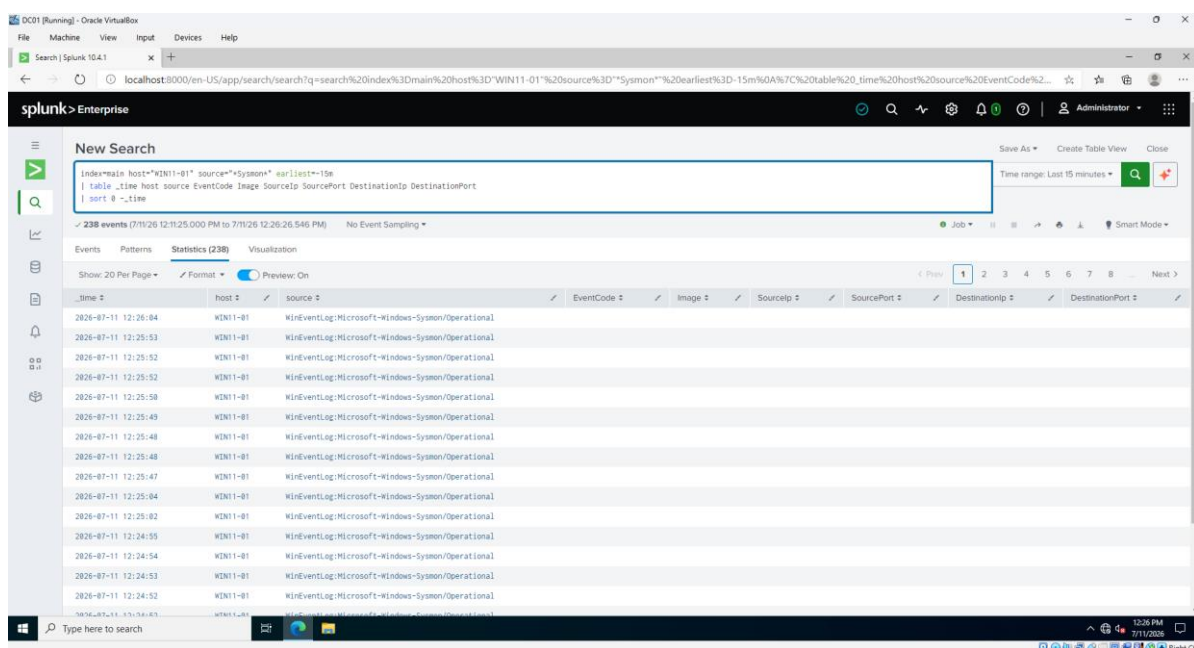


Figure 3. Fresh Sysmon events arrived after the Universal Forwarder input was reloaded.

PHASE 6

Phase 6 — Kali Linux Attacker Workstation

Objective

Prepare an isolated system for authorized reconnaissance and authentication testing.

Work Performed

- Imported the Kali Linux VirtualBox image and connected it to the Host-Only network.
- Assigned KALI-01 the static address 192.168.10.30.
- Verified communication with the Windows systems.
- Used Nmap for port scanning and NetExec for controlled SMB authentication testing.
- Stored attack and detection screenshots in organized phase folders.

Validation and Result

- Nmap activity generated searchable monitoring results and visualizations.
- NetExec generated the authentication sequence investigated in Phase 9.
- The same Kali system was used for the Phase 10 post-hardening validation.

Recorded Evidence

- Kali network screenshots, Nmap screenshots, and NetExec validation screenshots.

**AUTHORIZATION
BOUNDARY**

KALI-01 was used only within the isolated lab and only against systems owned by the project.

PHASE 7

Phase 7 — Safe Attack Simulations

Objective

Generate controlled activity that can be detected and investigated through the SIEM.

Work Performed

- Performed Nmap scans against the lab endpoint and reviewed event-code statistics, time charts, and port-scan timelines in Splunk.
- Performed repeated SMB authentication attempts with NetExec against WIN11-01.
- Used the localadmin account in the Phase 9 scenario and generated both failed and successful logons.
- Repeated local-authentication testing after hardening in Phase 10.

Validation and Result

- Nmap detection screenshots and an SOC monitoring dashboard were created.
- NetExec activity generated 31 failed logons and four successful localadmin network logons in the Phase 9 evidence window.
- The final Phase 10 NetExec test returned STATUS_ACCOUNT_LOCKED_OUT.

Recorded Evidence

- 04-Nmap-Detection-Statistics-byEventcode.png
- 04-Nmap-Detection-Statistics-byTimechart.png
- 08-Splunk-Full-Scan-Detection.png
- 09-Port-Scan-Timeline.png
- 21-NetExec-Blocked.png

SCOPE NOTE

The original blueprint also proposed encoded PowerShell and new-local-user simulations. They were not present in the submitted evidence and are therefore recorded as future extensions, not completed work.

PHASE 8

Phase 8 — SPL Detections, Alerting, and Dashboard

Objective

Convert collected telemetry into repeatable SOC monitoring and detection content.

Work Performed

- Created basic Nmap/port-scan detection searches and visualizations.
- Created authentication searches for Event IDs 4625 and 4624 by source address, account, host, and time.
- Developed a correlation approach for repeated failures followed by successful authentication.
- Enabled the Brute Force Detection alert.
- Built and verified an SOC monitoring dashboard.

Validation and Result

- The alert was shown enabled during Phase 10.
- The dashboard remained healthy after Windows and account-policy changes.
- Searches supported both the Phase 9 incident reconstruction and Phase 10 after-hardening validation.

Recorded Evidence

- Nmap detection queries and statistics screenshots.
- Brute Force Detection alert enabled.
- SOC monitoring dashboard healthy.

EVIDENCE IMPROVEMENT

A future portfolio update should add one screenshot of the alert firing, including trigger time, source IP, account, and drill-down results.

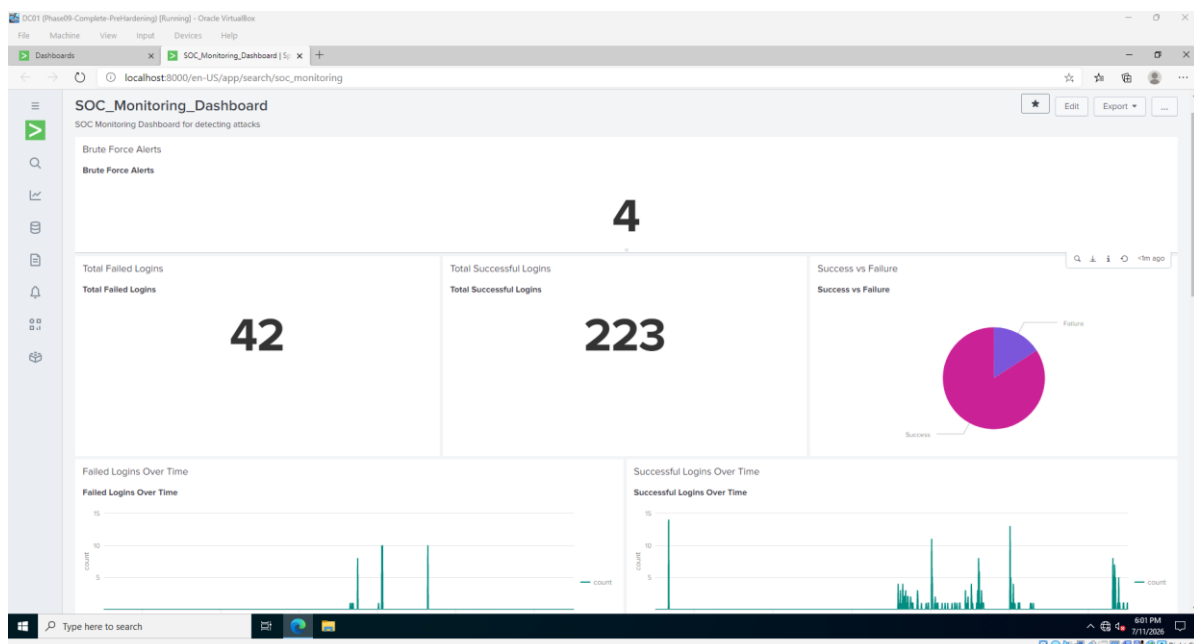


Figure 4. SOC monitoring dashboard remained operational after hardening.

PHASE 9

Phase 9 — Brute-Force Incident Investigation

Objective

Investigate the authentication attack, reconstruct the timeline, determine impact, restore missing telemetry, and document the incident.

Work Performed

- Identified 192.168.10.30 as the source of 31 failed logons against localadmin on WIN11-01.
- Confirmed four successful Event ID 4624 network logons from the same address and account.
- Validated Logon Type 3, consistent with remote network authentication.
- Excluded ANONYMOUS LOGON records from the success conclusion.
- Investigated the missing Sysmon Event ID 3 results, validated events locally, restarted the forwarder, and extracted XML fields.
- Produced and closed a professional high-severity incident report.

Validation and Result

- Incident finding: controlled credentials were successfully used against WIN11-01.
- Observed impact: successful network authentication to localadmin.
- Not demonstrated: persistence, privilege escalation, destructive activity, or data exfiltration.
- Phase 9 status: Closed — Lab Exercise.

Recorded Evidence

- 31 Event ID 4625 failures.
- Four Event ID 4624 successes.
- Sysmon Event ID 3 local and Splunk validation.
- Final Phase 9 incident report.

SEVERITY

HIGH — successful authentication followed repeated failed attempts from the same source address.

Time	Event	Finding
06:35–06:58	4625	31 failures from 192.168.10.30 against localadmin
06:46:41	4624	First successful localadmin network logon; Type 3
06:50:30	4624	Second successful network logon
06:58:40	4624	Third successful network logon
10:34:55	4624	Fourth successful network logon
12:17 / 12:25	Sysmon 3	Controlled telemetry connections to 192.168.10.10:9997

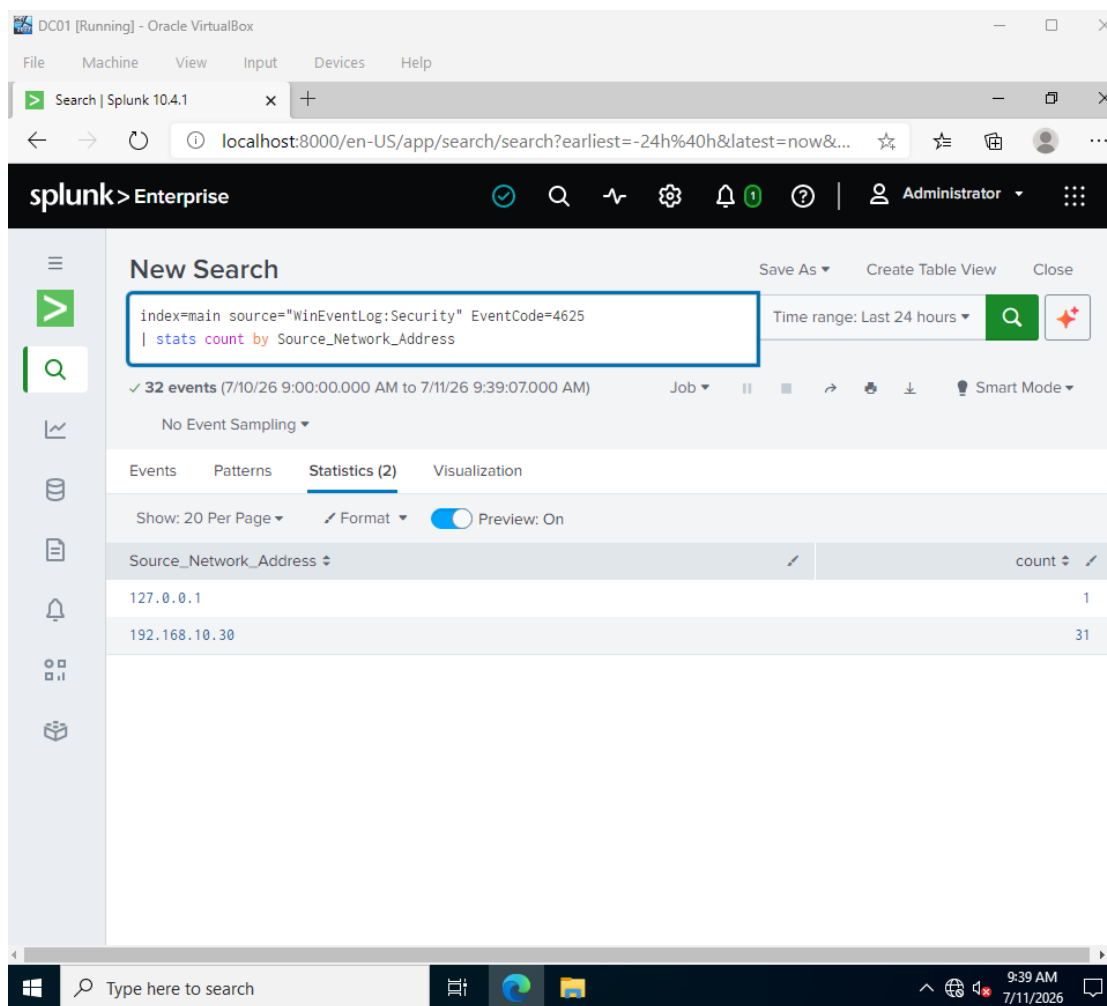


Figure 5. Failed logons grouped by source address; 192.168.10.30 generated 31 failures.

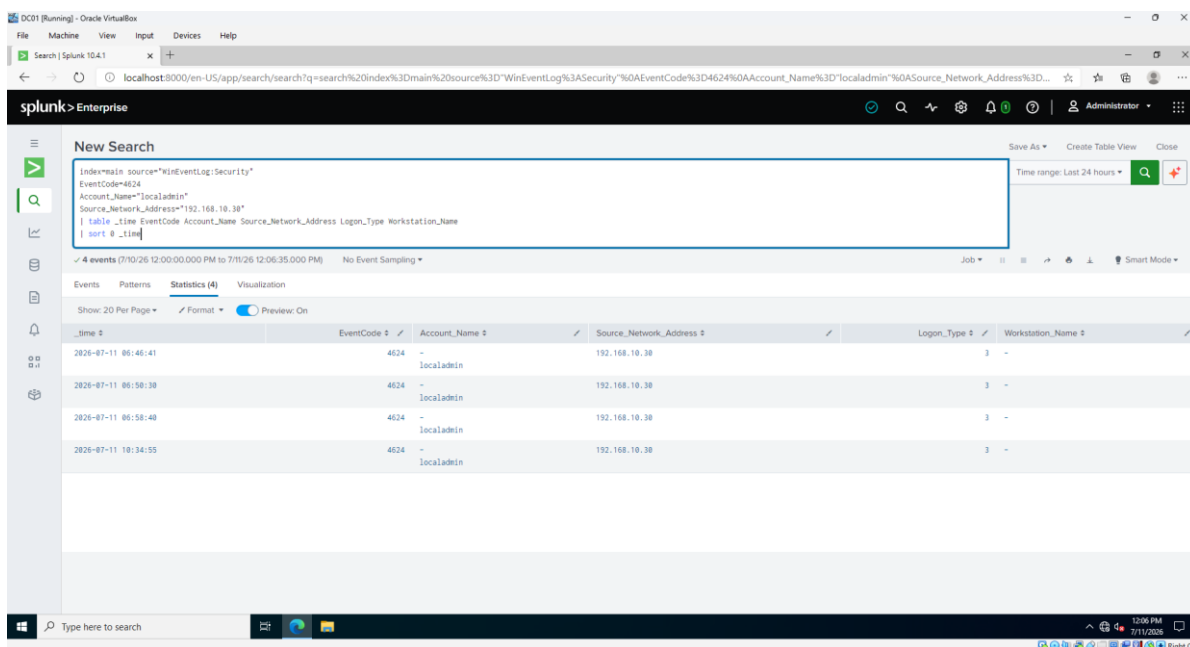


Figure 6. Four successful localadmin network logons from the attacker address.

PHASE 10

Phase 10 — Security Hardening and Validation

Objective

Reduce the likelihood of another successful credential attack and prove that preventive and detective controls remained effective.

Work Performed

- Created recovery snapshots for DC01, WIN11-01, and KALI-01 before changing security policy.
- Captured domain, local account, and SMB baseline configurations.
- Set domain and local lockout thresholds to five attempts with 15-minute duration and observation windows.
- Verified SMB1 disabled, SMB2 enabled, and SMB server signing required.
- Disabled RDP and verified all Windows Firewall profiles enabled with inbound traffic blocked by default.
- Verified Sysmon, SplunkForwarder, log ingestion, alert, and dashboard health.
- Repeated the NetExec authentication test and reviewed lockout and post-hardening evidence.

Validation and Result

- NetExec returned STATUS_ACCOUNT_LOCKED_OUT and did not display a successful-authentication marker.
- Splunk recorded post-hardening Event ID 4625 failures and an Event ID 4740 lockout event.
- No later successful localadmin Event ID 4624 appeared after the hardening window.
- Phase 10 status: PASSED and Closed — Lab Exercise.

Recorded Evidence

- Before/after policy screenshots.
- Host and protocol hardening evidence.
- Service and telemetry validation.
- Final blocked-attack and Splunk validation screenshots.

VALIDATION SCOPE

Phase 9 targeted localadmin. The final NetExec screenshot targeted the local Administrator account. This proves enforcement of local lockout, but a strict same-account replay remains a recommended improvement.

Control	Before	After	Assessment
Domain lockout	Threshold 0	5 attempts / 15 min	Improved
Local lockout	Never locked	5 attempts / 15 min	Improved
SMB1	Disabled	Disabled	Already compliant
SMB signing	Required	Required	Verified
RDP	Not baselined	Disabled	Exposure reduced
Firewall	Not baselined	All profiles ON	Verified
Logging	Operational	Operational	Revalidated
Attack outcome	Authentication succeeded	Account locked	Blocked


```

(kali@kali)-[~]
$ nxc smb 192.168.10.20 -u Administrator -p passwords.txt --local-auth
SMB 192.168.10.20 445 WIN11-01 [*] Windows 11 / Server 2025 Build 26100 x64 (name:WIN11-01) (d
omain:WIN11-01) (signing:True) (SMBv1:None)
SMB 192.168.10.20 445 WIN11-01 [-] Connection Error: The NETBIOS connection with the remote ho
st timed out.
SMB 192.168.10.20 445 WIN11-01 [-] Connection Error: The NETBIOS connection with the remote ho
st timed out.
SMB 192.168.10.20 445 WIN11-01 [-] Connection Error: The NETBIOS connection with the remote ho
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SMB 192.168.10.20 445 WIN11-01 [-] Connection Error: The NETBIOS connection with the remote ho
st timed out.
SMB 192.168.10.20 445 WIN11-01 [-] WIN11-01\Administrator:@X@50455766@X@ STATUS_ACCOUNT_LOCKED
_OUT
SMB 192.168.10.20 445 WIN11-01 [-] Connection Error: The NETBIOS connection with the remote ho
st timed out.
(kali@kali)-[~]

```

Figure 7. Post-hardening NetExec validation returned STATUS_ACCOUNT_LOCKED_OUT.

The screenshot shows the Splunk Enterprise interface with a search query: `index=main source="WinEventLog:Security" EventCode=4625 Account_Name="localadmin" | table _time Source_Network_Address Workstation_Name Failure_Reason Status Sub_Status | sort _time`. The search results show 25 events from 7/11/26 6:00:00.000 PM to 7/11/26 7:00:53.000 PM. The table below represents the data shown in the screenshot:

_time	Source_Network_Address	Workstation_Name	Failure_Reason	Status	Sub_Status
2026-07-11 18:28:26	192.168.10.30	-	Unknown user name or bad password.	0xC000006D	0xC000006A
2026-07-11 18:28:24	192.168.10.30	-	Unknown user name or bad password.	0xC000006D	0xC000006A
2026-07-11 18:28:22	192.168.10.30	-	Unknown user name or bad password.	0xC000006D	0xC000006A
2026-07-11 18:28:20	192.168.10.30	-	Unknown user name or bad password.	0xC000006D	0xC000006A
2026-07-11 18:28:18	192.168.10.30	-	Unknown user name or bad password.	0xC000006D	0xC000006A
2026-07-11 18:08:22	192.168.10.30	-	Unknown user name or bad password.	0xC000006D	0xC0000064
2026-07-11 18:08:20	192.168.10.30	-	Unknown user name or bad password.	0xC000006D	0xC0000064
2026-07-11 18:08:18	192.168.10.30	-	Unknown user name or bad password.	0xC000006D	0xC0000064
2026-07-11 18:08:16	192.168.10.30	-	Unknown user name or bad password.	0xC000006D	0xC0000064
2026-07-11 18:08:13	192.168.10.30	-	Unknown user name or bad password.	0xC000006D	0xC0000064
2026-07-11 18:08:11	192.168.10.30	-	Unknown user name or bad password.	0xC000006D	0xC0000064
2026-07-11 18:08:09	192.168.10.30	-	Unknown user name or bad password.	0xC000006D	0xC0000064

Figure 8. Failed authentication events remained visible in Splunk after hardening.

Key Splunk Searches

Successful localadmin network logons from the attacker address

```
index=main source="WinEventLog:Security" EventCode=4624
Account_Name="localadmin" Source_Network_Address="192.168.10.30"
| table _time EventCode Account_Name Source_Network_Address Logon_Type
| sort 0 _time
```

Failed authentication activity

```
index=main source="WinEventLog:Security" EventCode=4625
| stats count BY Source_Network_Address Account_Name host
| sort -count
```

Sysmon Event ID 3 extraction from XML

```
index=main host="WIN11-01" source="*Sysmon*"
| rex field=_raw "<EventID[^>]*>(?!<SysmonEventCode>\d+)</EventID>"
| search SysmonEventCode=3
| rex field=_raw "<Data Name=[\"]Image[\"]>(?!<Image>[^\"]+)"
| rex field=_raw "<Data Name=[\"]SourceIp[\"]>(?!<SourceIp>[^\"]+)"
| rex field=_raw "<Data Name=[\"]DestinationIp[\"]>(?!<DestinationIp>[^\"]+)"
| rex field=_raw "<Data Name=[\"]DestinationPort[\"]>(?!<DestinationPort>[^\"]+)"
| table _time SysmonEventCode Image SourceIp DestinationIp DestinationPort
```

Recommended failures-followed-by-success correlation

```
index=main source="WinEventLog:Security" (EventCode=4625 OR EventCode=4624)
| stats count(eval(EventCode=4625)) AS failures
count(eval(EventCode=4624)) AS successes
earliest(_time) AS first_seen latest(_time) AS last_seen
BY Source_Network_Address Account_Name
| where failures>=5 AND successes>=1
| convert ctime(first_seen) ctime(last_seen)
```

Account-lockout validation

```
index=main source="WinEventLog:Security" EventCode=4740
| table _time host TargetUserName CallerComputerName
| sort -_time
```

Troubleshooting Record and Lessons Learned

Problem	Diagnosis	Resolution / lesson
VM performance and storage	VM files initially located on slower HDD	Moved working VM files to SSD while preserving configuration
Host-only communication	Wrong or inactive adapter caused connectivity problems	Returned systems to Host-Only Adapter #2 and verified pings
Sysmon Event 3 returned zero	Local Sysmon contained Event 3, so logging was not the root cause	Restarted SplunkForwarder, reloaded input, and extracted XML fields
Sysmon fields appeared empty	renderXml=true stored required values inside _raw XML	Used rex extraction for EventID and network fields
Initial Phase 10 NetExec timeouts	Some attempts did not complete SMB authentication	Continued controlled validation until Windows returned account locked out
Blueprint versus implementation	Separate indexes were planned but main was used	Document the working configuration as authoritative

Security Lessons

- Detection alone does not prevent access; preventive controls such as lockout and restricted remote administration are also required.
- A successful authentication event must be interpreted carefully and correlated by source, account, host, logon type, and time.
- Analysts should not overstate evidence. Authentication success did not prove execution, persistence, privilege escalation, or data theft.
- Telemetry health must be validated locally and centrally before concluding that an event did not occur.
- Post-hardening attack repetition provides stronger evidence than configuration screenshots alone.

Residual Risks and Recommended Improvements

Priority	Item	Recommended action
High	Minimum password length remains seven	Increase to at least 14 characters and use unique administrative credentials
High	Final NetExec account differs from Phase 9 account	Repeat the validation against localadmin for a strict before/after comparison
Medium	Default ADMIN\$, C\$, and IPC\$ shares remain	Restrict access to approved management hosts or an administration subnet
Medium	Event 4740 timing is not tied to one final command sequence	Correlate attack, 4625, 4740, and absence of 4624 in one time window
Medium	Alert enabled but trigger evidence is limited	Capture the alert firing and preserve its result details
Low	Encoded PowerShell simulation not evidenced	Add a harmless encoded-command test with Sysmon Event 1 detection
Low	New-local-user simulation not evidenced	Add an authorized test and detect Windows Event ID 4720
Portfolio	Earlier-phase images are not consolidated	Create phase folders and a sanitized public README with selected evidence

MITRE ATT&CK Context

Technique	ID	Observed context
Brute Force	T1110	Repeated credential attempts against a local Windows account
Valid Accounts	T1078	Successful use of tested credentials during the Phase 9 lab scenario
Network Service Scanning	T1046	Nmap reconnaissance against the isolated Windows endpoint

PUBLIC- RELEASE CHECK

Before publishing, sanitize passwords, attempted credential strings, personal account names, and any information that should not be publicly exposed.

Evidence Register

The following evidence groups support the project record. Phase 9 and Phase 10 filenames correspond to the submitted packages; Phase 0–8 items are referenced in the exported conversation.

Phase	Evidence group	Examples
0	Preparation	VirtualBox, Host-Only adapter, DHCP disabled, project folders
1	DC01 deployment	VM settings, Windows Server installation, boot/network screenshots
2	AD and DNS	Domain services, user/account, domain-policy evidence
3	WIN11-01	VM, static IP, domain use, connectivity evidence
4	Splunk	Web interface, receiver, searches and indexed events
5	Sysmon / Forwarder	Local Event 3, inputs.conf, service state, recovered ingestion
6	Kali	Static IP, communication, Nmap and NetExec evidence
7	Simulations	Nmap statistics, scan timeline, brute-force results
8	Detection content	Port-scan searches, brute-force alert, SOC dashboard
9	Investigation package	Figures 1–11 and Phase 9 Incident Report
10	Hardening package	Snapshots, baselines, policy changes, telemetry, blocked validation

Primary Phase 9 Evidence

ID	Evidence	Purpose
01–03	Failed logons, attacker IP, targeted account	Authentication-source and target proof
04, 06, 07	Timeline and exploratory correlation	Investigation context
08	Successful login after brute force	Primary compromise evidence
09–11	Local Sysmon, forwarding restored, Event 3 extraction	Telemetry validation

Primary Phase 10 Evidence

ID	Evidence	Purpose
01–04	Snapshots and pre-hardening baselines	Recovery and comparison
05–13	Policy, SMB, RDP, firewall, shares	Hardening and residual exposure
14, 16–20	Services, logs, alert, dashboard	Monitoring health
21	NetExec blocked	Final attack validation
23–25	4625, 4740, no later 4624	Post-hardening SIEM proof

Final Conclusion

The SOC Home Lab achieved its core technical objective: a complete, evidence-based defensive workflow operating inside an isolated environment. The project progressed from virtualization and identity infrastructure through endpoint telemetry, attack simulation, SIEM detection, incident investigation, security hardening, and validation.

Phase 9 proved that repeated failed authentication attempts were followed by successful localadmin network logons. Phase 10 converted the investigation findings into preventive controls, preserved recovery points, revalidated monitoring, and demonstrated account-lockout enforcement during the final test.

The remaining items are improvements rather than blockers. The strongest next steps are an exact localadmin before/after replay, a captured alert trigger, additional safe simulations for encoded PowerShell and local-user creation, and a sanitized public README that consolidates the earlier phase evidence.

FINAL STATUS

CORE SOC HOME LAB COMPLETE — build, attack, detect, investigate, harden, and validate objectives demonstrated.